The Bekenstein Sphere: Finite Capacity, the Hellinger Little Sphere, and the Boundary-Localized Residual

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Abstract

The Bekenstein bound is the dimensionful clothing of a fact the categorical-simplex framework already owns.

The Bekenstein bound states that a finite region of space holding finite energy holds at most finitely much information, $S \le 2\pi kRE/(\hbar c)$, with both sides — written as $S/k \le 2\pi RE/(\hbar c)$ — pure numbers. We show that Paper 32's one functional $\Phi = \log Z$ contains this statement at its dimensionless core, and that the "little sphere" of the bound is *literal*, not analogical. Four results, all firewall-clean (Paper 32 §12) and all backed by deterministic exact-arithmetic audits:

1. **[P] The state space is a little sphere.** The categorical row R, $Q = |R|$, embeds through the Hellinger map $p \mapsto 2\sqrt{p}$ as a *sector of a round sphere of constant curvature* $K = \frac{1}{4}$ *(radius* 2*)*, dimension $D = Q - 1$, occupying solid-angle fraction exactly $1/2^Q$ — the structural integer of Paper 32 §13.2. We verify $K = \frac{1}{4}$ symbolically (identically in coordinates) and the sector fraction by independent Monte-Carlo.
2. **[P] Capacity = the finite Bekenstein bound, saturated at the centroid.** The information capacity is $S_{\max} = \log Q = \log(\dim)$, attained at the centroid $p_* = (1/Q, \dots, 1/Q)$ — the maximally-mixed/thermal state. It is the $\beta = 0$ face of the finite Bekenstein inequality $S(\rho) - S(\sigma) \le \beta(\langle H\rangle_\rho - \langle H\rangle_\sigma)$, which is the Fenchel–Young inequality of Φ (Paper 54, Theorem 4); so the bound is an internal relation of Φ in an exact sense, not a comparison of two independent objects. The sphere and the capacity themselves are classical (Čencov; Amari–Nagaoka; Shannon). The Fisher–Rao "size" of the region stays bounded ($R_F < \pi$, diameter $= \pi$ exactly) while $S_{\max} = \log Q$ grows — the strong-holographic signature: *information grows, geometric size does not.*
3. **[P, cited] Why exactly $\hbar$ and c, and not G.** The dimensionless core of the bound is Casini's inequality $\Delta S \le \Delta\langle K\rangle$ for the modular Hamiltonian K of the vacuum, a KMS state (§6). For a Rindler wedge, Bisognano–Wichmann give $K = 2\pi \int x\, T_{00}\, d^3x/(\hbar c)$: $\hbar c$ converts the boost generator into dimensionless modular energy, and 2π is the KMS inverse temperature. G enters only at black-hole saturation, where the entropy is read as a *dimensionful area* — permanently firewall-external (Paper 32 Principle 12.1). The Planck area $\ell_P^2 = \hbar G/c^3$ is then the framework's **information cell** read in dimensionful units: the framework fixes the holographic pixel *count* (the conductor lattice, a pure number) and disowns the pixel *size* ℓ_P^2 (§10.1).
4. **[A]/[P-structural] Boundary-localization on the arithmetic (unit-group) boundary — not a spatial boundary, not a geometric area.** The closure residual is supported on the conductor-fusion boundary $M = (\mathbb{Z}/N)^\times$ (Paper 32 Cor 5.1d.3). A ladder audit ($N \in \{6, 30, 210, 2310\}$) reproduces §6's 18/20 at $N = 6$ and establishes a ladder-wide parity theorem: $T_{\text{unit}} = 0$, the boundary is cubically *self-inert* — **the gravitating residual is boundary-supported but bulk-sourced**. The dimensional boundary share $\varphi(N)/(N-1) \to \prod_{p|N}(1 - 1/p)$ is *number-theoretic*, not a geometric codimension-1 area exponent. A third audit settles what kind of object the boundary is: the geometric area-law negative *survives the Fisher→Bures metric upgrade* (the metric is isotropic, $g_{p_*} = Q\, I_H$, at the saturating centroid, so no metric reweights it), yet under the framework's intrinsic *conductor* (arithmetic-mass) weighting the boundary carries a **non-vanishing** share $\to \prod_p (1 - \frac{1}{p^2-p+1}) \approx 0.515$. The boundary is arithmetic mass, not geometric area; that limit has the firewall-clean closed form $\zeta(6)/(\zeta(2)\zeta(3)) = 2\pi^4/(315\,\zeta(3))$ (§11).

The Bekenstein bound is not a new physical input: it is relative-entropy positivity against a KMS state (Casini; Paper 54), and its $\beta = 0$ face is the framework's capacity $\log Q$. Its dimensionless core is exact; its dimensionful magnitude (and the area law $S = A/4G$) is permanently external by the same firewall principle.

Keywords: Bekenstein bound, holographic principle, Fisher–Rao geometry, Hellinger embedding, categorical effective action, conductor packets, relative entropy, KMS state, modular Hamiltonian, dimensional firewall, finite-dimensional entropy bound.

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1. Scope and claim status

This companion uses Paper 32's three-tier convention verbatim ([P] proven given Papers 13/17; [A] structural pending a named finite audit; [C] conjectural, reduced) and adds **[F]** for firewall-binding exclusions. It introduces no new substrate: every object below is a reading of Paper 32's $\Phi = \log Z$, its cumulant tower, its conductor packets, and its firewall principle. The eight finite audits of §8 are run and archived (`empirical/paper32_1_audit1_capacity_law.py` through `…_audit8_beta_limit.py`); all are pure `numpy`/`sympy`/`mpmath`, the deterministic gate.

Foundation (Paper 54). Theorem 1 (Φ as the generating function of the completely passive states; Fenchel–Young gap $= D(\rho\|p_\theta)$) and Theorem 4 (the finite Bekenstein inequality is passivity of a KMS state; Casini's continuum form). **Dependency spine (Paper 32).** Thm 3.1 (space = mass = second cumulant, i.e. the static FDT of Paper 54 Thm 2); §5 + Cor 5.1d.3 (residual supported on M); §6 thread T4 (the 18/20 count); §9.3 (the c- and $\hbar$-conversion faces — now an analogy only, §5); §9.7 (the projection is a monotone contraction); §9.9 (Raychaudhuri localization); §13.1 (positioning against emergent gravity); §13.2 (the structural integer $1/2^Q$); Rem 16.2b ($K = \frac{1}{4}$, scalar $R = D(D-1)/4$, exact-certified); §12 (the firewall principle, true by construction).

2. The one functional and the finite region

Let R be a finite modulus, $Q = |R|$, Δ_R° the open categorical simplex, $H_R = \{u : \sum u_a = 0\}$ its tangent hyperplane, and $\Phi(\theta) = \log Z(\theta)$ the log-partition. Paper 32 reads the cumulant tower of Φ as the four objects physics separates only because it presupposes a manifold; here only two are needed:

$$\nabla^{(0)}\Phi \;\leftrightarrow\; \text{entropy / occupation (the } \log Z \text{ at the vacuum)}, \qquad \nabla^{(2)}\Phi = g \;=\; \text{Fisher metric} \;=\; \text{space} \;=\; \text{inertia}.$$

The Bekenstein bound compares *a region of space* with *the information in it*. In ordinary physics these are two things. Here, by **Theorem 3.1 (Paper 32)**, the region's geometry *is* the second cumulant of the same functional whose zeroth-order content is the entropy. So:

Proposition 2.1 (The bound is internal). [P, via Paper 54, Theorems 1 and 4] *The finite Bekenstein inequality $S(\rho) - S(\sigma) \le \beta(\langle H\rangle_\rho - \langle H\rangle_\sigma)$, $\sigma = p_\theta$, $\theta = -\beta H$, is the Fenchel–Young inequality of Φ and its Legendre dual $-S$: its defect is $\Phi(\theta) + \Phi^*(\rho) - \langle\theta, \rho\rangle = D(\rho\|p_\theta)$. Both sides are values of one functional and its conjugate; there is no second ruler to calibrate against the first.*

Draft 0.1 phrased this as order-0 content against order-2 content ("Fisher volume = space"). The order-2 reading is the static fluctuation–dissipation theorem (Paper 54, Theorem 2) and is not needed for the inequality. Casini's continuum derivation (§6) uses exactly the same inequality, for the modular Hamiltonian, so this is shared ground, not an advantage over it.

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3. The little sphere, made literal — [P]

"This little sphere" is not a figure of speech. Under the Hellinger map $\phi(p) = 2\sqrt{p}$, the simplex Δ_R° maps onto a region of the Euclidean sphere $\{x \in \mathbb{R}^Q : |x| = 2\}$, because $\sum_a (2\sqrt{p_a})^2 = 4\sum_a p_a = 4$. The Fisher–Rao metric is exactly the pullback of the round metric:

$$g_p(u,u) = \sum_a \frac{u_a^2}{p_a} = \sum_a \Big(\frac{u_a}{\sqrt{p_a}}\Big)^2 = \big|d\phi(u)\big|^2, \qquad u \in H_R.$$

Theorem 3.1 (The state space is a constant-curvature little sphere). [P] *The categorical state space is a geodesically convex* ***sector of the round sphere of radius*** 2*, hence of constant sectional curvature $K = \frac{1}{4}$ and scalar curvature $R = D(D-1)/4$, $D = Q - 1$. The sector is the all-positive orthant: of the 2^Q sign-orthants of the ambient sphere exactly one is compatible with the positive cone, so the sector occupies solid-angle fraction exactly $1/2^Q$ — Paper 32 §13.2's admissible structural integer.*

Proof. The pullback identity above is exact; a Euclidean sphere of radius $r = 2$ has constant sectional curvature $1/r^2 = \frac{1}{4}$, and a D-manifold of constant sectional curvature κ has scalar curvature $D(D-1)\kappa$. The orthant count is combinatorial. ■

Audit anchor (§8.1). Audit 1 computes the Gaussian curvature of the $Q = 3$ row *symbolically* and returns $K(x,y) = \frac{1}{4}$ **identically** in coordinates; verifies $|\,2\sqrt{p}\,|^2 = 4$ and the Fisher = pullback identity to machine ε on $Q \in \{3, 4, 6, 12, 30\}$; and corroborates the sector fraction $1/2^Q$ by isotropic Monte-Carlo (e.g. $Q = 6$: exact 0.015625 vs MC 0.01556). The little sphere is literal.

Figure 1 assembles the whole arc this paper traces — the sphere born as a point, gaining a dimension per distinction, shrinking its pixel toward a finite floor (§§3–12).

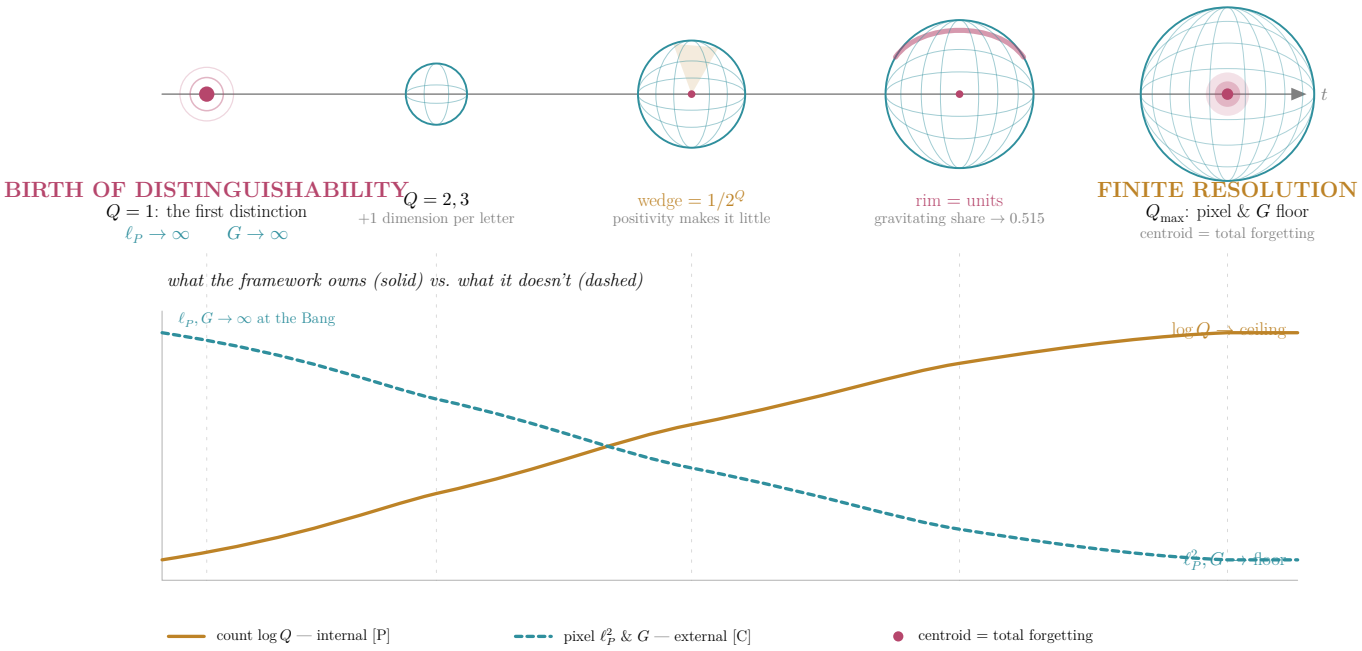


Figure 1: **The little sphere through time.** The categorical state space is a round sphere of curvature $K = \frac{1}{4}$. It is born as a single point at the Big Bang ($Q = 1$, the first distinction; in the pixel gauge $\ell_P \to \infty$, $G \to \infty$), gains a dimension per added distinguishable state as the count $\log Q$ grows (solid; internal, owned), and shrinks its Planck-pixel toward a finite floor (dashed; external) at the centroid — maximal entropy, the saturating "total-forgetting" state. The shaded wedge is the positive orthant ($1/2^Q$); the highlighted rim is the unit boundary whose conductor share $\to \zeta(6)/(\zeta(2)\zeta(3)) \approx 0.515$. Cosmological readings (§12) are [C], fenced.

The radius (2) and curvature ($\frac{1}{4}$) are **Q-independent pure numbers**; what grows with the row is the *dimension* $D = Q - 1$ and the *capacity* (§4). The sector fraction $1/2^Q$ is what makes the sphere *little*: the physical region is an exponentially small wedge of the full sphere.

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4. Capacity is the finite Bekenstein bound, saturated at the centroid — [P]

For a system with a Q-state categorical description (equivalently a dim $= Q$ lift $D(\mathbb{C}^Q)$), the maximum entropy is

$$S_{\max} = \log Q = \log(\dim),$$

attained uniquely at the centroid $p_* = (1/Q, \ldots, 1/Q)$ — the maximally-mixed state, the infinite-temperature/thermal fixed point of the modular flow (Paper 32 §9.5, Connes–Rovelli). This is precisely the Bekenstein bound for a finite-dimensional Hilbert space: *a bounded region admits at most* $\log(\dim)$ *nats.*

Theorem 4.1 (Capacity law). **[P]** *The categorical capacity is $S_{\max} = \log Q$, a pure number (the log of the structural integer Q), saturated at the centroid. The saturating state is the maximal-Fisher-volume point and the metric maximum $g_{p_*} = Q\, I_H$.*

Theorem 4.2 (Strong-holographic signature). **[P]** *The Fisher–Rao size of the region is bounded independently of Q: the centroid-to-vertex distance is $R_F(Q) = 2\arccos(1/\sqrt{Q}) \nearrow \pi$, and the vertex-to-vertex diameter is exactly π for every $Q \geq 2$. Hence as the row grows the capacity $S_{\max} = \log Q \to \infty$ while the geometric size stays $< \pi$: information grows, the little sphere does not.*

Audit anchor (§8.1). Audit 1 confirms $S_{\max}$ = centroid entropy $= \log Q$ to machine ε across $Q \in \{3, \ldots, 210\}$, and $\max_Q R_F = 3.0035 < \pi$ while the capacity rises $1.099 \to 5.347$.

Provenance. Theorems 3.1, 4.1 and 4.2 are classical: the Hellinger embedding and curvature $\frac{1}{4}$ (Čencov 1972; Amari–Nagaoka 2000), the Fisher–Rao distance $2\arccos\sum\sqrt{p_i q_i}$ and diameter π (Atkinson–Mitchell 1981), and the capacity $\log Q$ (Shannon 1948); see Paper 54, Proposition 5.2. What this paper adds is their exact certification and their use: $\log Q$ is the $\beta = 0$ face of the Bekenstein inequality (§6).

Remark 4.3 (Two energies, one capacity — demoted 2026-10-01: circular). *Status: descriptive only.* The "entropic energy" below is $E_{\text{sat}} = S_{\max}/(2\pi R_F)$, i.e. it is defined so that the bound saturates, so it cannot support any claim about which energy the bound binds. The non-circular answer is Proposition 6.1: the bound binds the modular energy $\beta\Delta\langle H\rangle$ (in the continuum $\Delta\langle K\rangle$) of a KMS reference state. The original remark follows. The framework carries two distinct scales: the **inertial** scale $E_g = Q$ (the second-cumulant/Fisher eigenvalue at the centroid, Thm 3.1's "mass") and the **entropic** scale $\sim \log Q$ that pairs with capacity. The geometric Bekenstein product $2\pi R_F E_g \sim 2\pi^2 Q$ over-saturates: $S_{\max}/(2\pi R_F E_g) \to 0$. The capacity binds to the entropic energy $E_{\text{sat}} = S_{\max}/(2\pi R_F) \sim \log Q$, not to the inertial Q. The centroid saturates $S = \log(\dim)$ directly. *(This is a descriptive readout, not a pass/fail; it locates the Bekenstein energy as the entropic, not the inertial, scale.)*

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5. Why the bound carries $\hbar$ and c, and not G — [P, cited]

Write the bound dimensionless: $S/k \le 2\pi\, RE/(\hbar c)$ — pure number $\le$ pure number, exactly the category Paper 32 §12 permits. Its exact dimensionless core is Casini's inequality $\Delta S \le \Delta\langle K\rangle$ (§6), in which the modular Hamiltonian $K = -\log\sigma_V$ of the vacuum is a pure-number operator. For a Rindler wedge, Bisognano–Wichmann (1975, 1976) give

$$K = \frac{2\pi}{\hbar c}\int_{x>0} x\, T_{00}\, d^3x.$$

The integral is a boost generator, with units of energy × length; $\hbar c$ converts it into dimensionless modular energy, and 2π is the KMS inverse temperature of the vacuum with respect to boosts.

Proposition 5.1 (Constant content of the bound). [P, cited] *The Bekenstein bound compares an entropy with a modular energy. Written in laboratory units through Bisognano–Wichmann, it needs exactly $\hbar$ and c, in the combination $\hbar c$, together with the KMS factor 2π. G is absent from the inequality and enters only at black-hole saturation (§10), when the entropy is identified with a horizon area $A/4G$ — a curvature conversion the firewall externalizes.*

The clean three-way ledger:

object	conversion needed	tier
capacity $\log Q$	none	**[P]** internal ($\beta = 0$ face, §6)
Bekenstein RHS $2\pi RE/\hbar c$	$\hbar c$ (Bisognano–Wichmann) and 2π (KMS)	**[P, cited]**; the step $\Delta\langle K\rangle \le 2\pi RE/\hbar c$ is [A]
black-hole area $A/4G$	G (curvature conversion)	**[F]** firewall-external

Superseded reading. Draft 0.1 derived the $\{\hbar, c\}$ content from the two "conversion faces of the single Bures lift" (Paper 32 §9.3) at tier [P-adjacent]. It survives only as an analogy [A]; the load-bearing reason is Bisognano–Wichmann.

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6. The inequality is passivity of a KMS state; saturation is the reference state — [P]

Draft 0.1 read the "$\leq$" as the data-processing contraction of Paper 32 Cor 9.7. The load-bearing inequality is sharper and different in kind: positivity of relative entropy against a KMS reference state (Paper 54 §6).

Proposition 6.1 (The "$\leq$" is passivity of a KMS state). [P] *Let $\sigma = e^{-\beta H}/Z$ be a faithful Gibbs state, $\beta \geq 0$. For every state ρ,*

$$S(\rho) - S(\sigma) \;\leq\; \beta(\langle H\rangle_\rho - \langle H\rangle_\sigma),$$

with equality iff $\rho = \sigma$; the defect is $D(\rho\|\sigma)$ (Paper 54, Theorem 4). At $\beta = 0$ ($\sigma = \mathbf{1}/Q$) this is the capacity law of Theorem 4.1, $S(\rho) \leq \log Q$, with defect $D(\rho\|\mathbf{1}/Q) = \log Q - S(\rho)$: saturation occurs exactly at the centroid. On the unitary orbit of σ it is passivity: no work can be extracted from a Gibbs state.

The continuum form [P, cited]. For a region of a quantum field theory with vacuum-reduced state $\sigma_V = e^{-K}/Z$, Casini (2008) shows $\Delta S \leq \Delta\langle K\rangle$ for every state; both differences are finite although each term diverges. For the Rindler wedge K is 2π times the boost generator (§5), and for a positive-energy excitation within proper distance R of the edge $\Delta\langle K\rangle \leq 2\pi RE/(\hbar c)$, which is Bekenstein's form. This step is [A]: Casini derives it for weakly self-gravitating systems, semiclassically. **The 2π of the bound is the KMS inverse temperature of the vacuum for boosts [P, cited]**, replacing draft 0.1's "[C], character/Hopf-angle candidate".

Relation to data processing. Positivity of relative entropy is the trivial instance of data processing (the channel to a point), so draft 0.1's reading is not false, but the content lies elsewhere: in the reference state being KMS. The arrow of Paper 32 Cor 9.7 (monotone contraction under projection) and the bound are two different uses of relative entropy, not "the same monotone contraction". Audit anchor: Paper 54 certificate K5 (exact signs on 44 classical and 8 quantum rational pairs; the $\beta = 0$ and isentropic faces).

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7. Boundary-localization on the arithmetic (unit-group) boundary — [A]/[P-structural], with an honest area-law negative

The "binding" information — the part the spatial (inertial) projector cannot keep closed, the gravity-adjacent residual of Paper 32 §5 — is the closure residual T_{cross}, and **Cor 5.1d.3** proves it is supported *exactly* on the conductor-fusion boundary $M = (\mathbb{Z}/N)^\times$ (the units / primitive characters), sharply. This is the arithmetic analogue [A] of the holographic feature: *the residual lives on the unit-group boundary, not in the bulk of the conductor poset* (and Raychaudhuri focusing localizes onto the same $T^{(4)}|_M$, Paper 32 §9.9).

Which boundary. $M = (\mathbb{Z}/N)^\times$ is the unit group, the top packet of the conductor (divisor) poset — an *arithmetic* boundary. It is not a spatial boundary, and nothing in this section localizes information on a surface in space; "holographic" and "area" below are used only as analogies [A].

A ladder audit ($N \in \{6, 30, 210, 2310\}$, the primorials $P_2 ... P_5$) makes this exact and surfaces a new structural theorem.

Proposition 7.1 (Boundary parity-inertness; the residual is bulk-sourced). [P] *At every even modulus N — in particular every squarefree primorial ≥ 6 — the units packet is cubically self-inert: no triple of units closes,*

$$u_1, u_2, u_3 \in (\mathbb{Z}/N)^\times \implies u_1 + u_2 + u_3 \not\equiv 0 \pmod N, \qquad \text{hence } T_{\text{unit}} = 0.$$

Proof. N even $\Rightarrow$ every unit is odd (coprime to 2); a sum of three odd numbers is odd; and 0 mod N is even. ■

This is the ladder-wide generalization of Paper 32 Lemma 6.1(L1) ($3+3+3 \equiv 3 \not\equiv 0$ at $\mathbb{Z}/6$): the residual is **boundary-supported but bulk-sourced** — the boundary cannot cubically self-source; every closing

cubic vertex touching the boundary also carries a bulk leg. In gravitational language [A, analogy]: the boundary *carries* the residual but the bulk *sources* it.

The area-law fork, settled (matched pair). - **HORN A (geometric area law):** the boundary share follows a codimension-1 area exponent. - **HORN B (number-theoretic):** the dimensional boundary share is $\beta_{\dim}(N) = \varphi(N)/(N-1) \to \prod_{p|N}(1-1/p)$, set by Euler's φ, with no geometric area exponent.

Audit result (§8.2): HORN B. $\beta_{\dim}$ decreases $0.4000 \to 0.2759 \to 0.2297 \to 0.2079$ along the ladder, tracking $\prod(1-1/p)$ to $< 5 \times 10^{-3}$. Boundary-localization of the residual is sharp and [P] (Cor 5.1d.3, Prop 7.1); the *quantitative geometric area law is not reproduced*. In the §16.4 register, the negative is the contribution: it draws the boundary between what the framework supplies (sharp arithmetic boundary-localization) and what it does not (a geometric area exponent), so the holographic match cannot be overread.

Measure dependence: the negative is geometric, the positive is arithmetic — [P]. Is HORN B an artifact of *how* the boundary is measured? Audit 3 (§8.3) tests two pre-registered rescue horns and settles what kind of object the holographic boundary is.

- **HORN A′ (metric rescue) — REFUTED.** At the Bekenstein-saturating centroid $\rho_* = I/Q$ the Fisher metric is isotropic, $g_{p_*} = Q\, I_H$ (Paper 32 §5.3), and the SLD quantum-Fisher / Bures metric *coincides with it* on the diagonal packet directions ($g_Q/g_F = 1$ to machine ε, eigenvalue Q on every packet at $N = 6, 30$). The boundary/bulk *volume* ratio is therefore metric-independent and equals the dimension count: **no metric — classical Fisher or quantum Bures — rescues the area law at the saturating state.**
- **HORN B′ (arithmetic rescue) — CONFIRMED.** Weight each mode by its intrinsic *conductor* $c(k) = N/\gcd(k, N)$ — the packet-defining arithmetic mass (Paper 32 §5.1), the framework's energy analog, hence the natural Bekenstein weight (Bekenstein weights information by energy). The boundary share is $\varphi(N)\, N \big/ \sum_{c|N,\, c>1} \varphi(c)\, c$, with **exact** values $\frac{3}{5}$, $\frac{6}{11}$, $\frac{5040}{9481}$, $\frac{277200}{526223}$ = 0.600, 0.545, 0.532, 0.527 along the ladder, converging to

$$\beta_{\text{cond}}(\infty) = \prod_p \Big(1 - \frac{1}{p^2 - p + 1}\Big) \approx 0.515 \;\neq\; 0.$$

  The conductor-weighted *cubic coupling* is even more boundary-heavy: $0.95 \to 0.90 \to 0.89 \to 0.88$.

So the holographic boundary is **not a metric/geometric area** (no metric makes it non-vanishing at the saturating centroid) but **is a non-vanishing arithmetic locus** ($\sim$ half the conductor-weighted measure, ~ 0.88 of the conductor-weighted cubic coupling). Audit 2's vanishing dimensional share was a counting artifact; the firewall-clean positive statement is *arithmetic* — the conductor is a pure number (Paper 32 §12). *(Forced-match guard, Paper 35: the conductor weighting is pre-justified as the packet-defining arithmetic mass and reported alongside the negative dimension/metric measures, not selected to manufacture a result.)*

Texture (descriptive). The cubic coupling is *over-weighted* on the boundary relative to its dimension but is not boundary-dominated: the fraction of closing triples carrying a boundary leg falls $0.90 \to 0.62 \to 0.53 \to 0.49$ across the ladder, settling near $\frac{1}{2}$ while the boundary occupies only $\sim 21\%$ of the dimensions at $N = 2310$. The base row $\mathbb{Z}/6$ (90%) is the most boundary-concentrated; large primorials split the cubic coupling roughly evenly between bulk and boundary.

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8. The eight finite audits

All eight are pre-registered, run, and archived. Firewall discipline (Paper 32 §12): only pure numbers / exact integer counts and their rational ratios are emitted — no length, energy, $\hbar$, c, or G.

8.1 Audit 1 — the capacity law (`paper32_1_audit1_capacity_law.py`)

Question. Across the ladder $Q = |R|$: is the capacity the finite Bekenstein bound $S \leq \log(\dim)$ saturated at the centroid; is the state space literally a $K = \frac{1}{4}$ sector of fraction $1/2^Q$; does capacity grow while Fisher size stays bounded?

Verdicts (all PASS). - **V1** capacity = centroid entropy = $\log Q$ to machine ε (saturated). - **V2** little sphere literal: $K = \frac{1}{4}$ **symbolically identical**; sector fraction $1/2^Q$ exact, MC-corroborated. - **V3** $R_F < \pi$ (max 3.0035) while capacity rises $1.099 \to 5.347$ — strong-holographic. - **V4** *(descriptive)* energy split: inertial scale Q, Bekenstein-saturating scale $\sim \log Q$, ratio $\to 0$.

8.2 Audit 2 — the boundary-information fraction (`paper32_1_audit2_boundary_fraction.py`)

Question. Is the residual boundary-localized, and does the boundary share obey a geometric area law (HORN A) or a number-theoretic law (HORN B)? Regression: reproduce Paper 32 §6 (T4)'s 18/20 at $N = 6$.

Verdicts. - **V1 (regression) PASS.** $N = 6$: $T_{\text{all}} = 20$, pure-bulk = 2, $\beta_{\text{cubic}} = 18/20 = 0.9$, $T_{\text{unit}} = 0$ — exactly Paper 32 §6 T4. - **V2 (Prop 7.1).** $T_{\text{unit}} = 0$ at all of $N \in \{6, 30, 210, 2310\}$: boundary cubically self-inert ladder-wide; residual bulk-sourced. - **V3 (fork) HORN B CONFIRMED.** $\beta_{\text{dim}} = \varphi(N)/(N-1)$ decreases $0.40 \to 0.21$, tracking $\prod(1 - 1/p)$: number-theoretic, not a geometric area exponent. - **V4 (texture).** β_{cubic}: $0.90 \to 0.62 \to 0.53 \to 0.49$ — over-weighted on the boundary, settling near $\frac{1}{2}$.

N	$\varphi(N)$	dim	β_{dim}	T_{all}	T_{bulk}	T_{unit}	β_{cubic}
6	2	5	0.4000	20	2	0	0.9000
30	8	29	0.2759	812	308	0	0.6207
210	48	209	0.2297	43 472	20 288	0	0.5333
2310	480	2309	0.2079	5 329 172	2 695 412	0	0.4942

8.3 Audit 3 — measure dependence (`paper32_1_audit3_measure_dependence.py`)

Question. Is Audit 2's area-law negative an artifact of the measure? Two pre-registered rescue horns: upgrade the metric (HORN A′, Fisher → Bures), or weight by the intrinsic conductor mass (HORN B′).

Verdicts. - **A′ REFUTED.** At $\rho_* = I/Q$: Fisher $= Q\, I_H$ isotropic (coefficient spread $< 10^{-14}$, eigenvalue Q on every packet); SLD/Bures coincides on the diagonal packet directions ($g_Q/g_F = 1$). The boundary/bulk volume ratio is metric-independent = dimension count $\to 0$. - **B′ CONFIRMED.** Conductor share $\frac{3}{5} \to \frac{6}{11} \to \frac{5040}{9481} \to \frac{277200}{526223} = 0.600 \to 0.527$, converging to $\prod_p(1-1/(p^2-p+1)) \approx 0.515 \neq 0$; conductor-weighted cubic share $0.95 \to 0.88$.

Net: the negative is geometric (no metric reweights the boundary at the saturating state), the positive is arithmetic (the conductor mass puts $\sim$ half the weight on the boundary) — both firewall-clean.

8.4 Audit 4 — the zeta-ratio closed form (`paper32_1_audit4_zeta_ratio.py`)

Question. Does the conductor-share limit (Audit 3, HORN B′) have a closed form?

Verdict. The per-prime identity $(p^2-p)/(p^2-p+1) = (1-p^{-2})/(1+p^{-3})$ is verified symbolically; the closed forms $\zeta(6)/(\zeta(2)\zeta(3))$ and $2\pi^4/(315\,\zeta(3))$ agree to 25 digits ($= 0.5145101015\ldots$); the truncated product converges to them from above (0.5145101364 at $P = 10^6$, residual $\sim 3 \times 10^{-8}$). **Zeta-ratio closed form confirmed (Thm 11.1).**

8.5 Audit 5 — the constant out there (`paper32_1_audit5_totient_constant.py`)

Question. Is $\zeta(6)/(\zeta(2)\zeta(3))$ a known constant?

Verdict. Its reciprocal $\zeta(2)\zeta(3)/\zeta(6) = 1.9435964\ldots$ is **Landau's totient constant** (OEIS A082695): the totient-sieved mean $\frac{1}{x}\sum_{n\le x} n/\varphi(n) \to 1.943589$ at $x = 10^6$ (error 7×10^{-6}), and the Euler product $\prod_p(1 + 1/(p(p-1))) = 1.9435963$ cross-checks. The Bekenstein boundary constant is $1/\langle n/\varphi(n)\rangle$ (Remark 11.2). **Landau-totient identification confirmed.**

8.6 Audit 6 — the 5-joining transition (`paper32_1_audit6_prime5_transition.py`)

Verdict. The 6→30 step is an **atomic baton-change**, not a swell: packets 3→7, gravity boundary relabels conductor 6→30 at $t_* = 0.05$, cross-coupling ×40, $T_{\text{unit}} = 0$. The logit appears only under a weighting temperature, with sweet spot $\beta_* : 0.495 \to 0.823$. Feeds §13.1.

8.7 Audit 7 — the prime-joining ladder (`paper32_1_audit7_prime_ladder.py`)

Verdict. Closed step-laws verified against brute force (packets $= 2^k - 1$, cross-couplings $= (N-1)(N-2)$): the 7-step and 11-step computed exactly, 13+ by formula; β_* flow $0.495 \to 0.906$; horizon handoff $t_* = 1/(p(p-1))$ sharpening. Feeds §13.

8.8 Audit 8 — the sweet-spot limit (`paper32_1_audit8_beta_limit.py`)

Question. Does the sweet-spot exponent $\beta_*(P_k)$ reach 1, or converge to a computable limit below it?

Verdict. $\beta_\infty < 1$**, pinned.** The finite condition reduces (squarefree multiplicativity) to the Euler-product equation $(\star)$ $\prod_p(1 + 1/((p-1)p^\beta)) = 2$; root-found at 40-digit precision (prime-zeta tail; validated by $\log L(1) = \log[\zeta(2)\zeta(3)/\zeta(6)]$ to 10^{-16}), $\beta_\infty = 0.954352187581\ldots$, stable across head cutoffs $X = 2\times10^4 \to 2\times10^5$. The ladder converges **monotonically from below** ($\beta_*(P_{78}) = 0.9536$); `mpmath.identify` finds **no elementary closed form** — a new constant in Landau's Φ_6 Euler-product family. Closes §13.3.

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9. Firewall ledger and falsifiers

Ledger. - **[P]:** little sphere literal ($K = \frac{1}{4}$, $r = 2$, fraction $1/2^Q$, $\dim Q - 1$); capacity $\log Q$ saturated at the centroid; size bounded while capacity grows; space = information = $\nabla^{(2)}\Phi$ (internal relation, Prop 2.1); the "$\le$" = relative-entropy positivity against a KMS state, with $\log Q$ its $\beta = 0$ face (Prop 6.1; Paper 54 Thm 4); residual boundary-localized (Cor 5.1d.3); boundary parity-inertness, residual bulk-sourced (Prop 7.1); the geometric area-law negative is robust to the Fisher→Bures upgrade (metric isotropic at ρ_*) while the conductor-weighted boundary share is non-vanishing, $\to \prod_p(1-1/(p^2-p+1)) \approx 0.515$ (Audit 3). - **[P, cited]:** the $\{\hbar, c\}$-not-G content of the bound and its 2π (Prop 5.1; Bisognano–Wichmann; Casini). **[A]:** the step $\Delta\langle K\rangle \le 2\pi RE/\hbar c$ (semiclassical). - **[A]:** the holographic reading of boundary-localization on the arithmetic (unit-group) boundary. (The energy-split of Remark 4.3 is demoted: circular.) - **[C], reduced:** the geometric area-law exponent (settled-negative as HORN B here, §16.4 register); (the 2π coefficient moved to [P, cited]: Bisognano–Wichmann KMS inverse temperature). - **[F] firewall-binding:** the dimensionful bound $2\pi kRE/(\hbar c)$ and the area law $S = A/4G$ as **magnitudes**; the value of G; any length in metres or energy in joules. Permanently external by Paper 32 Principle 12.1 — identical status to $c \equiv 1$.

Falsifiers. - **B1.** A measured time-translation residual off the boundary M (forbidden by Cor 5.1d.3) ⇒ §7 collapses. - **B2.** A pure-unit cubic self-coupling ($T_{\text{unit}} \neq 0$) at any even N ⇒ refutes Prop 7.1 (impossible by parity). - **B3.** $\beta_{\dim}(N)$ following a geometric codimension-1 exponent rather than $\prod(1-1/p)$ ⇒ would promote HORN A; refuted by the audit. - **B4.** Any assignment of a dimensionful value to the bound, the area law, or G ⇒ firewall violation (Paper 32 Principle 12.1). Existence, invariance, and pure-number content are the only admissible claims. - **B5.** A non-isotropic Fisher/Bures metric at the centroid ρ_*, or a conductor-weighted boundary share $\to 0$ ⇒ refutes Audit 3's geometric-negative / arithmetic-positive split (both impossible: $g_{p_*} = Q\, I_H$ exactly, and $\prod_p(1-1/(p^2-p+1)) > 0$).

10. Black-hole saturation needs G — [F]

Bekenstein saturation by a black hole promotes the boundary-localized residual to a *dimensionful horizon area*, $S = A/(4G)$ (in $\hbar = c = 1$ units). By Paper 32 §12.1(4) the projected curvature defect $\delta\mathcal{R}_{\text{proj}}$, although internal as a pure number on the conductor-fusion boundary $\partial_n S \subseteq \mathcal{D}_N$, carries G the instant it is read as an area. Hence:

Proposition 10.1 (Saturation is firewall-external). [F] *The framework gives the saturating state (centroid/maximally-mixed/thermal), the boundary-localization of the saturating information (Prop 7.1), and the KMS-positivity inequality (Prop 6.1) — all internal. It does not, and by Paper 32 Principle 12.1 cannot, give the area-law magnitude $A/4G$, the value of G, or any black-hole entropy in absolute units. The form is internal; the magnitude is external.*

This places the Bekenstein/black-hole reading exactly in Paper 32 §13.1's stance against the emergent-gravity school: the *information content* of the bound is arithmetized; the *thermodynamic-gravitational magnitude* (temperature, G, the area in m^2) is fenced off as requiring the operator-algebraic / continuum completion downstream.

10.1 The Planck area, in the framework's sense

The Planck area $\ell_P^2 = \hbar G/c^3$ is the conversion that turns information into area: the Bekenstein–Hawking law $S = A/(4\ell_P^2)$ assigns one unit of entropy to every four Planck areas. In the framework this object has a precise — and precisely external — status.

Proposition 10.2 (Planck area = the fully-dressed information cell). [F] *$\ell_P^2 = \hbar G/c^3$ is the simultaneous product of all three conversion constants the firewall externalizes: the two §9.3 Bures-lift faces c (space↔time) and $\hbar$ (Fisher-clock↔phase-clock), and the §9.6/§12.1(4) curvature conversion G. It is therefore the maximally firewall-external quantity — no path inside the finite substrate reaches it. What the framework owns is the dimensionless **count**: the holographic "pixels" are the cells of the conductor lattice $\mathcal{D}_N$ (the packets, the Q states, the capacity $\log Q$), a pure number. The framework fixes the pixel **count** and disowns the pixel **size** ℓ_P^2. "One bit per four Planck areas" is the dimensionful clothing of the framework's "one mode per conductor cell".*

So in Paper 32's sense the Planck area is not a length the framework computes but the *name of the cell-to-area conversion* — exactly the locus where information (internal, counted) becomes geometry (external, sized). The Compton-wavelength trap of Paper 32 §9.3/§12.3 is its sibling: both are firewall-external because both convert a pure-number count into a dimensionful length scale.

The factor $1/4$ — a guarded echo. [C], forced-match-flagged. The Bekenstein–Hawking $S = A/4$ (Planck units) carries the pure number $\frac{1}{4}$; the categorical little sphere carries Gaussian curvature exactly $K = \frac{1}{4}$ (§3, Thm 3.1). It is tempting to read the black-hole $\frac{1}{4}$ as the Hellinger curvature. **We decline the identification.** The area-law magnitude is firewall-external (Prop 10.1), so its coefficient is not an internal quantity; and Paper 32 §16.4 — the packet central-charge "6" that looked intrinsic and proved a terminated coincidence — is the cautionary precedent. The $\frac{1}{4}/K$ echo is recorded as flagged, not claimed: admissible to note, forbidden to assert.

11. The boundary constant is a zeta ratio — [P] identity, [A]/[C] reading

The conductor-weighted boundary share (§7, HORN B′) has an exact closed form. Because $p^3+1=(p+1)(p^2-p+1)$, every primorial factor telescopes,

$$\frac{p^2-p}{p^2-p+1}=\frac{1-p^{-2}}{1+p^{-3}},$$

so the infinite-primorial limit is a ratio of Riemann zeta values.

Theorem 11.1 (The boundary constant). [P]

$$\beta_{\mathrm{cond}}(\infty)=\prod_p\frac{1-p^{-2}}{1+p^{-3}}=\frac{\prod_p(1-p^{-2})}{\prod_p(1+p^{-3})}=\frac{1/\zeta(2)}{\zeta(3)/\zeta(6)}=\boxed{\frac{\zeta(6)}{\zeta(2)\,\zeta(3)}}=\frac{2\pi^4}{315\,\zeta(3)}=0.514\,510\,101\,508\ldots$$

Proof. The per-prime identity is elementary (Audit 4, verified symbolically); $\prod_p(1-p^{-s})=1/\zeta(s)$ and $\prod_p(1+p^{-3})=\zeta(3)/\zeta(6)$ are the standard Euler products. The numeric product converges to the closed form from above (Audit 4: 0.5145101364 at $P=10^6$, residual $\sim 3\times10^{-8}$ = the prime tail). ■

Reading (structural). [A]/[C], forced-match-guarded. Three features, each a clean fact, none claimed beyond the identity: 1. The leading factor is $1/\zeta(2)=6/\pi^2\approx0.6079$ — the **density of squarefree integers**. This is apt: the boundary $M=(\mathbb{Z}/N)^\times$ is the squarefree-maximal-conductor packet, so its asymptotic conductor share is led by the squarefree density and corrected downward by the cubic-order factor. 2. The Euler factors are (order 2) $\prod(1-p^{-2})$ over (order 3) $\prod(1+p^{-3})$: the boundary share is the **order-2 (Fisher/metric) density divided by the order-3 (Amari–Chentsov/cubic) density** — the Connection-F order-2/order-3 spine of Paper 32 §16, here as a zeta ratio. The numerator and denominator are themselves the canonical divisor-family limits of T33 §8.4: the smooth Jordan-totient ratios $\widetilde{J}_k/\mathrm{pri}^k\to1/\zeta(k)$ give $1/\zeta(2)$ (order 2) and, via $\prod_p(1+p^{-3})=\zeta(3)/\zeta(6)$, the order-3 factor — so the Bekenstein boundary constant is a ratio of two smooth-primorial emissions. 3. The exponents $\{2,3,6\}$ are the metric order, the cubic order, and the base conductor $6=2\cdot3$ — the same factorization that organizes Paper 32 throughout.

Forced-match guard (Paper 35; Paper 32 §16.4). Theorem 11.1 is an exact identity [P]; the reading is [A]/[C]. We do **not** claim the framework *derives* a zeta value, that the constant touches RH, or that the order-2/order-3 gloss is more than suggestive. The cautionary precedent is Paper 32 §16.4 — the packet central-charge "6" that looked intrinsic and proved to be a terminated division-algebra coincidence. Here the identity is rigorous; only its significance is held open. What is firewall-clean and settled: the Bekenstein boundary's conductor share is the pure number $\zeta(6)/(\zeta(2)\zeta(3))$.

Remark 11.2 (The constant, identified — Landau's totient constant). [P] identity; [A] aptness. The reciprocal $\zeta(2)\zeta(3)/\zeta(6)=1.943\,596\ldots$ is a classical named constant: **Landau's totient constant** (Landau 1900; OEIS A082695), with three standard faces,

$$\frac{\zeta(2)\zeta(3)}{\zeta(6)}=\lim_{x\to\infty}\frac{1}{x}\sum_{n\le x}\frac{n}{\varphi(n)}=\lim_{x\to\infty}\frac{1}{\log x}\sum_{n\le x}\frac{1}{\varphi(n)}=\sum_{k\ge1}\frac{\mu^2(k)}{k\,\varphi(k)}.$$

So the Bekenstein boundary constant is exactly the **reciprocal of the asymptotic mean of** $n/\varphi(n)$ (verified by totient sieve to $x=10^6$: mean = 1.943589, Audit 5). This is *structurally forced, not a numerical coincidence*: the conductor-fusion boundary $M=(\mathbb{Z}/N)^\times$ is the **unit group**, $|M|=\varphi(N)$, so any constant governing the units' asymptotic conductor share must be a totient-ratio constant — and the squarefree support $\mu^2(k)$ of the third face is the program's own squarefree-primorial substrate. (We checked and **rule out** the nearby average-subgroup-count constant of $\mathbb{Z}_m\times\mathbb{Z}_n$, which is $1/\zeta(2)$, not this — a honest near-miss declined.) Cyclotomically, each factor is $\Phi_6(p)/(p\,\Phi_1(p))$ with $\Phi_6(p)=p^2-p+1$ — the 6th cyclotomic polynomial, the base-conductor signature.

Remark 11.3 (Corpus recurrence — the weighting is the hypercharge construction). [observed] The conductor weighting that produces the constant is not new to this paper: it is the **same operation as the hypercharge** $Y = \frac{1}{2}P_2 - \frac{1}{3}P_3$ (Paper 32 §10.3, the "unique conductor-weighted sector contrast"), and as Paper 13 Def. 14.26's "conductor-weighted covariance." What is new here is taking its **primorial limit** — a conductor-weighted sector share pushed to the full primorial is precisely what converts the finite rational $\varphi(N)N/\sum_{c|N}\varphi(c)c$ into a zeta ratio. The construction is corpus-native; the zeta-ratio constant is its first limit.

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12. The pixel over time (speculative frontier — fenced)

This section is the paper's speculative frontier, fenced accordingly: every claim is [C] or external, the firewall (Paper 32 Principle 12.1) admits only dimensionless ratios, and each conjecture carries an explicit falsifier (GPD discipline, Paper 35). Nothing here is required for §§1–11.

12.1 The arrow is internal; the pixel size is external

The holographic dictionary $\log Q = A/(4\ell_P^2)$ (§4, §10) reads: the count (left) equals the area in pixels (right). Sort it by the firewall.

- The **count** $\log Q$ and its **growth** are *internal*. The second law here is not a postulate but the data-processing inequality / monotone-metric contraction (Paper 32 Cor 9.7, [P]): coarse-graining loses distinguishability, so the readable count can only grow. *The arrow is owned.*
- The **pixel size** $\ell_P^2 = \hbar G/c^3$ is *external* (§10.1, [F]): the framework fixes the count, never the size.

The internal arrow therefore forces $A/(4\ell_P^2)$ to grow. *That it grows* is owned; *how the world books the growth* is the single external freedom this section probes.

12.2 Two gauges for one arrow

Observation 12.1 (gauge freedom). [C] *The growth of $A/(4\ell_P^2)$ admits two external bookkeepings: the **area gauge** (ℓ_P^2 fixed, A grows — the horizon expands, mainstream holography) and the **pixel gauge** (extent held, ℓ_P^2 shrinks, $\ell_P^2 \propto 1/S$). Because the firewall owns only the gauge-invariant count, "more pixels" and "smaller pixels" are two external descriptions of one internal fact — the count grew. The framework does not select the gauge.*

12.3 The pixel-clock conjecture

Conjecture 12.2 (entropy-set pixel). [C], ratio-only. *In the pixel gauge the pixel tracks the entropy,*

$$\frac{\ell_P^2(t)}{\ell_P^2(t_0)} = \frac{S(t_0)}{S(t)}, \qquad \text{i.e.} \qquad \ell_P^2(t) \propto \frac{1}{S(t)}.$$

Admissible only as written — a ratio of dimensionless ratios (Paper 32 §12.2 admits ratios, never magnitudes). With $\hbar, c$ the fixed conversion faces (§5), the drift lives in the curvature conversion: $G(t) \propto \ell_P^2(t) \propto 1/S(t)$.

Three readings, each [C]: **(i)** ℓ_P is a clock — the size of a single distinction records how much the universe has forgotten. **(ii)** Gravity self-quenches: gravity is the residue of forgetting (§7), so in this gauge gravity drives entropy production, entropy growth shrinks the pixel, and a smaller pixel is a smaller G — gravity spends itself producing the entropy that dilutes it. **(iii)** G decreases monotonically with the arrow (subject to §12.5).

12.4 The finite-alphabet floor

The categorical alphabet is **finite**: $Q < \infty$, so $\log Q$ has a ceiling, and in the pixel gauge the pixel cannot shrink without bound:

$$\ell_P^2 \geq \ell_{P,\min}^2 = \frac{A}{4 \log Q_{\max}}, \qquad G \geq G_{\min} \propto \ell_{P,\min}^2.$$

Consequence 12.3 (finite end). [C] *A finite alphabet gives the cosmos a finite-resolution end state: the pixel bottoms out, G reaches a floor, and the drift toward de Sitter/heat-death terminates at a definite, maximally-but-finitely-distinguished state — the centroid (the saturating "total-forgetting" state of §4, §6) reached, not approached forever. This is the framework-native departure from eternal de Sitter; its content is the dimensionless finite* $\log Q_{\max}$.

12.5 The falsifier, and the firewall-clean residue

From $G \propto 1/S$, $\dot{G}/G = -\dot{S}/S$. If the pixel-setting entropy is the **cosmological-horizon** entropy, $\dot{S}/S \sim H$, giving $\dot{G}/G \sim -H \approx -7 \times 10^{-11}\,\mathrm{yr}^{-1}$ — about 10^2–$10^3\times$ the lunar-laser-ranging bound $|\dot{G}/G| \lesssim 10^{-13}\,\mathrm{yr}^{-1}$. **The horizon-entropy version is therefore observationally excluded;** Conjecture 12.2 survives only if the pixel-setting entropy has $\dot{S}/S \ll H$ today (e.g. the slowly-growing black-hole/matter reservoir, not the horizon). *Which entropy sets the grain is the open question on which the conjecture turns.*

Firewall-clean regardless is the **dimensionless ratio of the two clocks**, $(\dot{S}/S)/H \sim \mathcal{O}(1)$: the forgetting clock and the expansion clock are synchronized up to a pure number, both freezing as $H \to H_\Lambda$, $\dot{S}/S \to 0$ at the de Sitter end. The magnitude of H is external (a unit, 1/time); only such ratios are admissible.

12.6 Open directions

(a) The Hubble parameter and its tension. [C] H is not constant in time: $H(t)$ falls through cosmic history toward the constant H_Λ, and "the Hubble constant" H_0 is only its present value (spatially constant, temporally not). A drifting pixel/G calibrates the early-universe standard ruler — the recombination sound horizon $r_s \propto 1/\sqrt{G\rho}$ — with a *different* G than local measurements use, precisely the early-vs-late mismatch the Hubble tension is. Conjecture 12.2 thus lives in the same solution space as varying-constant / early-modification resolutions of the tension (constrained by BBN; a door, not a fix).

(b) The Big Bang as the birth of distinguishability. [C] Run the arrow backward: the count falls toward its minimum. At the extreme $Q \to 1$ — one undifferentiated state, $\log Q \to 0$, **no distinguishability at all**. The Big Bang is then the first distinction, $Q : 1 \to 2$ — the alphabet acquiring its first letter beyond one, the first "this $\neq$ that." In the pixel gauge $\ell_P^2 \propto 1/\log Q \to \infty$ and $G \to \infty$ as $Q \to 1$: the initial state is one infinite pixel of infinitely strong gravity — the singularity read as *maximal undifferentiation*, not maximal density. This recasts the low-entropy past hypothesis (Boltzmann, Penrose) structurally: the universe began at low entropy because it began at **low count** — distinguishability can only be built up, never started maxed. The asymmetry between an infinite-G, $Q \to 1$ past and the finite-resolution $Q_{\max}$ future (§12.4) *is* the arrow.

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13. The prime-joining ladder

Climbing the primorial ladder $\mathbb{Z}/6 \to \mathbb{Z}/30 \to \mathbb{Z}/210 \to \mathbb{Z}/2310$ — a new prime joining the row at each step — is governed by exact closed step-laws (Audits 6–7). All firewall-clean.

Per-row invariants.

N	k	packets	$\varphi(N)$	β_{dim}	β_{cond}	β_*	E	F	cross-couplings
6	2	3	2	0.400	0.600	0.495	5	15	20
30	3	7	8	0.276	0.545	0.823	10	55	812
210	4	15	48	0.230	0.532	0.886	17	153	43 472
2310	5	31	480	0.208	0.527	0.906	28	406	5 329 172

Step-laws (prime p joins). [P] - **packets** $= 2^k - 1$ — the conductor lattice is the Boolean algebra of prime-subsets; a prime *doubles* it. - **gravity boundary** $\varphi(P_k) = \prod(p-1)$; joining p multiplies it by $(p-1)$. - **cross-couplings** (ordered cubic closing triples) $= (N-1)(N-2) \sim N^2$ — verified against brute force; closed form reaches any row instantly (the $\mathbb{Z}/30030$ value 901,710,812, where the heavy spectral audit is memory-blocked). - **horizon handoff** $t_* = 1/(p(p-1))$: 0.050, 0.024, 0.0091, 0.0064 for $p = 5, 7, 11, 13$ — the new top conductor seizes the boundary ever faster; the baton-change *sharpens* with each prime. - **inner space** $E = \sum_{p|N} p$, $F = \frac{1}{2}E(E+1)$ $(= \Lambda^2 E \oplus E^*)$: $F = 15, 55, 153, 406$. ($F = 55$ at $\mathbb{Z}/30$ is the Paper 32 App. B anchor; E, F beyond are the same rule extrapolated, not separately audited.)

§13.1 The 5-step is a baton-change, not a swell (Audit 6). The structural step is *atomic* — 3→7 packets, the old gravity boundary (conductor 6) demoted to interior, the new boundary (conductor 30) seizing at $t_* = 0.05$. A smooth logit appears *only* under a weighting temperature β: the boundary share is a sigmoid $0 \to 1$ in β with a sweet spot β_* (boundary = bulk $= \frac{1}{2}$). This is the step-vs-logit duality of §12 made quantitative — **the step is the granular truth, the logit is the archimedean temperature's smoothing of it.**

§13.2 Two opposing trends, a sediment of horizons (Audit 7). Each prime *shapes less* per mode (Mertens weight $-\log(1-1/p)$: $0.22, 0.15, 0.095$) yet *connects more* (cross-couplings $\sim N^2$: $\times 40, \times 53, \times 122$): diminishing voice, thickening web. The demoted boundaries **stack like sediment** — $6, 30, 210, 2310$, each the gravity horizon of its era, buried by the next prime. Gravity stays bulk-sourced at every even row (Prop 7.1). The observed invariants (3+1, $\sin^2\theta_W = 3/8$) sit on the $2, 3$ floor, untouched up the ladder; each prime builds a higher, quieter, denser GUT/gravity wing.

§13.3 The sweet-spot temperature flows $\frac{1}{2} \to \beta_\infty = 0.9543521876 < 1$ (Audit 8) — [P]. The sweet spot rises $\beta_* = 0.495 \to 0.823 \to 0.886 \to 0.906 \to \cdots$, the base row $\mathbb{Z}/6$ being the outlier at $\beta_* \approx \frac{1}{2}$ (the amplitude/Hellinger exponent). By squarefree multiplicativity the finite condition (boundary share $= \frac{1}{2}$) reduces to

$$\prod_{p \le p_k} \left(1 + \tfrac{1}{(p-1)p^\beta}\right) = 2 + \frac{1}{\prod_{p \le p_k}(p-1)p^\beta},$$

so the limit $\beta_\infty = \lim_k \beta_*(P_k)$ solves the Euler-product equation

$$\boxed{\prod_p \left(1 + \tfrac{1}{(p-1)p^\beta}\right) = 2} \qquad (\star).$$

At $\beta = 1$ the left side is **exactly Landau's constant** $\zeta(2)\zeta(3)/\zeta(6) = 1.9436\ldots$ (§11, Audit 5), which is < 2; each factor decreases in β; hence $\beta_\infty < 1$. **Audit 8 pins it.** Root-finding $(\star)$ at 40-digit precision — head over the primes $\le 2 \times 10^5$, closed by a prime-zeta tail, the method validated against $\log L(1) = \log[\zeta(2)\zeta(3)/\zeta(6)]$ to 10^{-16} — gives

$$\beta_\infty = 0.954352187581\ldots \qquad (1 - \beta_\infty = 0.045647812\ldots),$$

a computable constant strictly below the Landau (conductor) weight $\beta = 1$. The finite ladder approaches it **monotonically from below**, $\beta_*(P_k) = 0.495, 0.823, 0.886, 0.906, 0.920, \ldots, 0.9536$ at $k = 78$, the deficit

being exactly the vanishing correction $E_k = 1/\prod_{p\le p_k}(p-1)p^\beta$. A closed-form search (`mpmath.identify` over $\{\log 2, \zeta(3), \gamma, \pi, ...\}$ and unrestricted) returns nothing: β_∞ is a **new constant** — the unique real root of $(\star)$, in the same Euler-product (cyclotomic Φ_6) family whose $\beta = 1$ member is Landau's totient constant. The smooth-primorial paper T33 §12.5 *homes* this family: its $\beta = 1$ endpoint is the Φ_6 divisor-family emission $\widetilde{\Psi}_6/\widetilde{P}_{2^-} \to \zeta(6)/(\zeta(2)\zeta(3))$ (the smooth interpolation of β_{cond} of §7/§11) and its $\beta = 0$ endpoint the gravity-localization unit density; it confirms β_∞ sits *interior* to the two and is not reducible to a divisor-order emission — classified by position, not closed form, exactly as found here. The sweet spot does not reach 1: the lone top boundary can never quite balance the exponentially growing bulk, even at maximal conductor over-weight.

§11 and §13.3 are one curve, read twice. The limiting boundary share is $S(\beta) = 1/L(\beta)$ with $L(\beta) = \prod_p (1 + 1/((p-1)p^\beta))$ (the finite $S_{P_k} = 1/(L_k - E_k) \to 1/L$). At the canonical conductor weight $\beta = 1$ this is

$$S(1) = \frac{1}{L(1)} = \frac{\zeta(6)}{\zeta(2)\zeta(3)} = 0.5145\ldots,$$

exactly the boundary constant of Theorem 11.1. So the two readings are coordinates of a single curve at its two distinguished points: §11 fixes the weight ($\beta = 1$) and reads the share (0.5145); §13.3 fixes the share ($\frac{1}{2}$) and reads the weight ($\beta_\infty = 0.9544$). The content is one statement — at its own natural weighting the framework already sits a hair on the boundary-majority side of equipartition (51.45%), and β_∞ is the small back-off $\beta : 1 \to 0.9544$ that brings boundary and bulk to exact balance. The Bekenstein boundary constant and the sweet-spot exponent are the same Euler product $L(\beta)$, evaluated where the share is the Landau reciprocal and where it is one-half.

§13.4 The dimensional corner: a sieve conservation law — [P]. The curve $S(\beta)$ has a third distinguished point, the dimensional weight $\beta = 0$, where $S_N(0) = \varphi(N)/(N-1)$ — asymptotically the **unit density**

$$\frac{\varphi(P_k)}{P_k} = \prod_{i=1}^{k}\left(1 - \tfrac{1}{p_i}\right) =: Q_k,$$

the fraction of residues coprime to P_k, equivalently the density of integers surviving the sieve by $p_1, \ldots, p_k$. This is the **HORN B** quantity of §7 (Audit 2): it vanishes, $Q_k \to \prod_p (1 - 1/p) = 0$. Its decay across the ladder is an exact telescoping conservation law — since $Q_{k-1} - Q_k = Q_{k-1}/p_k$,

$$\sum_{k=1}^{\infty} \frac{1}{p_k} \prod_{i=1}^{k-1}\left(1 - \tfrac{1}{p_i}\right) = \sum_{k\ge 1}(Q_{k-1} - Q_k) = 1 - \lim_k Q_k = 1.$$

Each prime p_k claims the share Q_{k-1}/p_k — the natural density of integers whose **least prime factor** is p_k — and the shares partition unity precisely because the fully-sifted set has density zero ($\sum_p 1/p = \infty$, Euler). This is the sieve-theoretic content of §13.2's receding boundary, made into a closed law: every unit of distinguishability is claimed by exactly one prime, none escaping. The boundary-share curve, read at its three distinguished weights:

weight β	limiting share $S(\beta)$	character
0 (dimensional)	1 (telescopes)	closed, elementary (Euler/Mertens)
$\beta_\infty \approx 0.9544$	$\frac{1}{2}$ (root of $(\star)$)	transcendental, new constant (§13.3)
1 (conductor)	$\zeta(6)/(\zeta(2)\zeta(3)) = 0.5145\ldots$	Landau's totient constant (§11)

Only the dimensional corner carries a closed conservation law. Its *value* ($= 1$) is free; the *rate* at which the tail Q_k vanishes is Mertens' third theorem, $\prod_{p\le x}(1 - 1/p) \sim e^{-\gamma}/\log x$, and the arithmetic depth lives not

in the identity — which is Euler-elementary — but in the **error term** of that asymptotic, equivalently the uniform level of distribution of the primes on primorial moduli.

This dimensional corner is exactly the program's **gravity-localization fraction** $\varphi(P_k)/P_k$ (the GG-ledger "gravity localizes to the multiplicative units"); its canonical analytic interpolation is supplied by the smooth-primorial paper T33 §12.3, which replaces the discrete $e^{-\gamma}/\log x$ by the smooth $e^{-\tilde{\gamma}_*}/\log \mathrm{Li}^{-1}(z)$ with $\tilde{\gamma}_* = 0.6119\ldots$ the *smooth Mertens constant* — the discrete-to-smooth gap $\tilde{\gamma}_* - \gamma = \int_2^\infty (\pi - \mathrm{Li})/(u(u-1))\,du \approx 0.0347$ being a named Riemann-error integral. Moreover the "error term / level of distribution" residual named here is the same $\sqrt{x}$-scale Riemann zero-sum that T33 Cor 4.2/§9A.5 identifies as the smooth-vs-exact primorial gap (RH $\Leftrightarrow$ that gap is $O(\sqrt{x}\log^2 x)$): the holographic boundary residual and the analytic interpolation residual are one object, and neither paper closes it. The firewall reading is exact: a pure-number conservation law whose leading order is free and whose error term is where the hard arithmetic sits.

Conclusion

The Bekenstein bound asks how much information fits in a little sphere of given energy. Paper 32 answers without importing new physics: the little sphere is the Hellinger image of the categorical row — a sector (fraction $1/2^Q$) of the round sphere of curvature $K = \frac{1}{4}$, radius 2, dimension $Q-1$; its capacity is $\log Q$, saturated at the centroid; the inequality is relative-entropy positivity against a KMS state, whose $\beta = 0$ face is that capacity (Paper 54); $\hbar c$ enters through Bisognano–Wichmann, with 2π the KMS inverse temperature, while G enters only at saturation and is permanently external. The gravitating information is boundary-localized and, by a one-line parity theorem, boundary-supported but bulk-sourced — and the boundary is an *arithmetic* locus, not a geometric area: the area-law negative survives the Fisher→Bures metric upgrade, while under the framework's own conductor weighting the boundary carries a non-vanishing ~ 0.515 share. All eight audits pass on the deterministic gate, the $N = 6$ regression reproduces Paper 32 §6 exactly, and not one dimensionful magnitude is emitted. The boundary's conductor share is the pure number $\zeta(6)/(\zeta(2)\zeta(3))$ — the reciprocal of Landau's totient constant, fitting because the boundary is the unit group — and the Planck area is named for what it is: the external size of the framework's internal information cell. The Bekenstein bound is passivity of a KMS state — its finite core exact, its $\beta = 0$ face the framework's own capacity, its dimensionful magnitude fenced. And §12 opens the one door the firewall leaves ajar: if the pixel is entropy-set, the arrow writes itself into $\ell_P(t)$ — gravity self-quenching toward a finite-alphabet floor, the Big Bang receding to the first distinction $Q : 1 \to 2$ — every step [C], every step fenced.

Data availability. All [P] results follow from Paper 32 (Papers 13/17 as cited) and the eight archived audits `empirical/paper32_1_audit1_capacity_law.py`, `empirical/paper32_1_audit2_boundary_fraction.py`, `empirical/paper32_1_audit3_measure_dependence.py`, `empirical/paper32_1_audit4_zeta_ratio.py`, `empirical/paper32_1_audit5_totient_constant.py`, `empirical/paper32_1_audit6_prime5_transition.py`, `empirical/paper32_1_audit7_prime_ladder.py`, and `empirical/paper32_1_audit8_beta_limit.py` (pure `numpy`/`sympy`/`mpmath`; exit 0; transcripts reproducible).

References

Inherits Paper 32's reference set; the directly load-bearing additions:

1. J. D. Bekenstein, *Universal upper bound on the entropy-to-energy ratio for bounded systems*, Phys. Rev. D **23** (1981) 287. (the bound $S \le 2\pi kRE/\hbar c$)
2. J. D. Bekenstein, *Entropy bounds and the second law for black holes*, Phys. Rev. D **27** (1983) 2262; and the finite-Hilbert form $S \le \log(\dim)$.

3. R. Bousso, *A covariant entropy conjecture*, JHEP **07** (1999) 004; *The holographic principle*, Rev. Mod. Phys. **74** (2002) 825. (boundary/area localization of entropy)
4. G. 't Hooft (gr-qc/9310026); L. Susskind, *The world as a hologram*, J. Math. Phys. **36** (1995) 6377. (holographic principle)
5. T. Jacobson, *Thermodynamics of spacetime: the Einstein equation of state*, Phys. Rev. Lett. **75** (1995) 1260. (the emergent-gravity lineage Paper 32 §13.1 positions against; Clausius $\delta Q = T\,dS$ on horizons)
6. A. Connes and C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation* (1994). (thermal/modular fixed point = the saturating centroid; Paper 32 §9.5)
7. E. Landau, *Über die zahlentheoretische Funktion $\varphi(n)$ und ihre Beziehung zum Goldbachschen Satz* (1900). (the constant $\zeta(2)\zeta(3)/\zeta(6)$: mean of $n/\varphi(n)$, leading constant of $\sum_{n\le x} 1/\varphi(n)$; §11 Rem 11.2)
8. OEIS Foundation, sequence **A082695**, *Decimal expansion of $\zeta(2)\zeta(3)/\zeta(6)$ (Landau's totient constant)*. (§11 Rem 11.2)
9. F. Mertens, *Ein Beitrag zur analytischen Zahlentheorie*, J. reine angew. Math. **78** (1874) 46–62. (the third theorem $\prod_{p\le x}(1-1/p) \sim e^{-\gamma}/\log x$ — the decay rate of the §13.4 sieve tail)
10. H. Casini, *Relative entropy and the Bekenstein bound*, Class. Quantum Grav. **25** (2008) 205021. (§5–§6: $\Delta S \le \Delta\langle K\rangle$)
11. J. J. Bisognano and E. H. Wichmann, J. Math. Phys. **16** (1975) 985; **17** (1976) 303. (§5–§6: the wedge modular Hamiltonian is 2π times the boost generator)
12. Paper 54, *The One Functional, Derived* (`54_one_functional_from_passivity.md`). (§2, §6: Fenchel–Young form; finite Bekenstein = passivity of a KMS state)
13. C. E. Shannon (1948); N. N. Čencov, *Statistical Decision Rules and Optimal Inference* (1972; AMS 1982); C. Atkinson and A. F. S. Mitchell, *Rao's distance measure*, Sankhyā A **43** (1981) 345; S. Amari and H. Nagaoka, *Methods of Information Geometry* (AMS 2000). (§3–§4 provenance)

Appendix A. Audit transcripts (verdict lines)

Audit 1 (`paper32_1_audit1_capacity_law.py`): `K == 1/4 identically: True`; capacity = centroid entropy = $\log Q$ (V1 PASS); sector fraction $1/2^Q$ MC-corroborated (V2 PASS); $\max R_F = 3.0035 < \pi$, capacity $1.099 \to 5.347$ (V3 PASS); energy split inertial Q vs entropic $\sim \log Q$ (V4). **ALL PRE-CLAIMED VERDICTS PASS.**

Audit 2 (`paper32_1_audit2_boundary_fraction.py`): $N = 6$ regression $T_{\text{all}} = 20$, pure-bulk = 2, $\beta_{\text{cubic}} = 0.9$, $T_{\text{unit}} = 0$ (V1 PASS, matches §6 T4); $T_{\text{unit}} = 0$ at all rows (V2, Prop 7.1); $\beta_{\text{dim}} = 0.40 \to 0.21$ tracking $\prod(1-1/p)$ (V3 HORN B CONFIRMED); $\beta_{\text{cubic}} = 0.90 \to 0.49$ (V4). **REGRESSION PASS + FORK SETTLED (HORN B).**

Audit 3 (`paper32_1_audit3_measure_dependence.py`): HORN A′ REFUTED — Fisher $= Q\,I_H$ at ρ_* (coefficient spread $< 10^{-14}$), SLD/Bures $g_Q/g_F = 1$ on every packet; HORN B′ CONFIRMED — conductor share $0.600 \to 0.527 \to \prod_p(1-1/(p^2-p+1)) \approx 0.515$, conductor-weighted cubic share $0.95 \to 0.88$. **A′ REFUTED + B′ CONFIRMED (verdict measure-dependent, as pre-claimed): the boundary is arithmetic mass, not geometric area.**

Audit 4 (`paper32_1_audit4_zeta_ratio.py`): per-prime identity $(p^2-p)/(p^2-p+1) = (1-p^{-2})/(1+p^{-3})$ symbolically `True`; $\zeta(6)/(\zeta(2)\zeta(3)) = 2\pi^4/(315\zeta(3)) = 0.5145101015\ldots$ (25-digit agreement); product $\to$ closed form from above. **ZETA-RATIO CLOSED FORM CONFIRMED (Thm 11.1).**

Audit 5 (`paper32_1_audit5_totient_constant.py`): reciprocal $\zeta(2)\zeta(3)/\zeta(6) = 1.9435964\ldots$ identified as Landau's totient constant (OEIS A082695); sieved mean $n/\varphi(n) \to 1.943589$ at $x = 10^6$; Euler product $\prod_p(1+1/(p(p-1)))$ cross-checks. **LANDAU-TOTIENT IDENTIFICATION CONFIRMED.**

Audit 6 (`paper32_1_audit6_prime5_transition.py`): $6 \to 30$ atomic — packets $3 \to 7$, boundary φ $2 \to 8$ ($\times(5-1)$), cross-couplings $20 \to 812$ ($\times 40$), $T_{\text{unit}} = 0$; weighting-temperature logit β_* : $0.4948 \to 0.8227$;

horizon handoff $t_* = 0.050$; Mertens shaping 5 weaker than $2, 3$. **SHARP BATON-CHANGE + TEMPERATURE-LOGIT.**

Audit 7 (`paper32_1_audit7_prime_ladder.py`): closed forms vs brute force `True` (packets $= 2^k - 1$, triples $= (N-1)(N-2)$); ladder $6 \to 30030$, β_* flow $0.495 \to 0.920$, $t_* = 1/(p(p-1)) = 0.050, 0.024, 0.0091, 0.0064$; two opposing trends (Mertens $\downarrow$, connectivity $\uparrow$); sediment of horizons. **LADDER CHARACTERIZED; 7-step exact, 11/13 projected.**

Audit 8 (`paper32_1_audit8_beta_limit.py`): finite solver reproduces Audit 7 brute β_* $(0.4948, 0.8227, 0.8859)$; method validated $\log L(1) - \log[\zeta(2)\zeta(3)/\zeta(6)] = 1.19 \times 10^{-16}$, $L(1) = 1.94359644 < 2$; root of $(\star)$ $\beta_\infty = 0.954352187581\ldots$ (log-residual 7.7×10^{-40}, stable $X = 2\times10^4 \to 2\times10^5$); ladder monotone from below to $\beta_*(P_{78}) = 0.9536$; `identify` = `None` (no closed form). β_∞ **PINNED** < 1**.**